Sobhan Salarian

Data Engineer & Data Scientist

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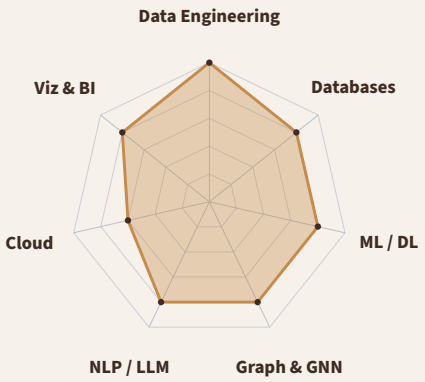
SKILLS OVERVIEW



Self-assessed proficiency, scale 1–5

TECHNICAL SKILLS

Programming: Python (Pandas, NumPy), SQL, HTML, CSS

Data Engineering: ETL pipeline development, PySpark, MapReduce (MrJob), Databricks (Delta Lake, Delta Live Tables, Unity Catalog, Asset Bundles)

Cloud: AWS (S3, EC2), Microsoft Azure (Azure SQL, Blob Storage)

Machine Learning / NLP: scikit-learn, PyTorch, Graph Neural Networks (GIN, GraphSAGE), Hugging Face (Transformers/Datasets), model evaluation (BLEU/chrF), LLM-agent simulation evaluation, LLM-as-judge (cross-provider),LLM agents (LangGraph), prompt-injection defence

Databases: MySQL, MongoDB (PyMongo), Neo4j (Cypher, Graph Data Science)

Visualization: Power BI, Matplotlib, Bokeh

Analysis: EDA, statistical analysis, feature engineering

Tools: Git, GitHub, Jupyter, VS Code, Linux/SSH (GPU servers), LaTeX, technical documentation

EDUCATION

Master of Data Science
Macquarie University, Sydney, NSW
2025–Dec 2026
WAM: 82.3. Highest mark in COMP6179 – Statistical Technologies for Data Science (Session 2, 2025)

Bachelor of Computer Engineering
Kermanshah University of Technology, Kermanshah, Iran
2018–2023

PROFILE

Data Engineer and Master of Data Science student, currently building tables and pipelines for a Synapse-to-Databricks Lakehouse migration at Cognitivo. Experienced across ETL, machine learning, and NLP, with two accepted papers on LLM-agent simulation reliability. President of Macquarie University's Big Data Society and Teaching Associate for COMP2270.

PROFESSIONAL EXPERIENCE

Work Experience

Teaching Associate – COMP2270 Networking and Cloud Computing, Macquarie University Aug 2026–Present

Teaching practical, workshop, and SGTA sessions supporting student learning in networking and cloud computing. Guide students through hands-on labs covering networking fundamentals and cloud infrastructure.

Data Engineer Intern – Cognitivo, Sydney, NSW Jul 2026–Present

- **Data Engineering:** Migrating business data and reporting from Microsoft Synapse to a Databricks Lakehouse. Built curated datasets, a star-schema model and Lakeview dashboards for a national general practice network, and deduplicated legacy ledger records into a validated revenue table. Building Asset Bundles and Unity Catalog pipelines that feed a client's Power BI reports directly.
- **AI Agents:** Added page awareness to the company's CRM AI assistant (LangGraph, TypeScript, MongoDB), with permission-scoped lookups and tested prompt-injection safeguards.

Database Specialist – IRISA Company, Isfahan, Iran Jul–Sep 2022

Cleaned and standardized datasets in Python, loading processed data into MySQL and collaborating with the reporting team on data requirements. Documented data processing steps and ran quality checks that improved pipeline reliability.

English Language Instructor – Kish Language School, Isfahan, Iran 2018–2020

Taught English to pre-intermediate and intermediate students aged 8–18, building strong communication and classroom management skills. Adapted teaching approaches across age groups and ability levels to improve participation and understanding.

Collaborations

President – Big Data Society (Website) 2026–Present

Lead a student data community founded by Prof. Amin Beheshti, bridging academia and industry through partnerships and workshops. Organize networking sessions and speaker events, collaborating with university stakeholders to drive engagement.

AI Hackathon Technical Designer – Macquarie University Representative (Hackathon Website) Jun 2026

Represented Macquarie University and the Big Data Society at an industry hackathon sponsored by NVIDIA, Gigabyte, LangChain, Cognitivo, UNSW, and Macquarie University. Retrieved and prepared hackathon datasets in JSON format to support fine-tuning and RAG implementation as part of the core design team.

PUBLICATIONS

Diagnosing the Believability–Reliability Gap in LLM-Agent Simulations of Online Communities 2026

International Conference on Web Information Systems Engineering (WISE 2026)
S. Mohseni, A. Beheshti, A. Shahsavandi, **S. Salarian**, U. Shahbaz. *Accepted.*

CERTIFICATIONS

Neo4j & Generative AI Fundamentals
Neo4j GraphAcademy • 2025
Generative AI, LLMs, and RAG/GraphRAG patterns; integrating Neo4j with generative AI and graph-based retrieval. Credential

Data Analysis Bootcamp
Daneshkar Academy • 2024
Statistics, SQL/MySQL, and data manipulation with Pandas/NumPy; dashboards in Power BI and Git/GitHub workflows.

LANGUAGES

Persian (Farsi)
Native

English
Professional Working Proficiency

A Believability–Reliability Dashboard: An Interactive Diagnostic Tool for LLM-Agent Social Simulations 2026
IEEE International Conference on Data Mining (ICDM 2026), Demo Track
S. Mohseni, **S. Salarian**, H. M. Abachi, U. Shahbaz, A. Beheshti. *Accepted.*

PROJECTS

Data Engineering & Big Data Projects

Graph-Based Fraud Detection with Neo4j and MongoDB 2026
Built a 50K-node transaction graph in Neo4j and applied Louvain, WCC and PageRank (Graph Data Science) to detect fraud rings, reaching 100% precision and 86.8% recall against ground-truth labels. Designed MongoDB aggregation pipelines and optimised query time from over 30s to under 1s.

Big Data Weather Analysis 2025
Processed large-scale weather datasets using MongoDB and MapReduce (MrJob), building distributed data pipelines for extreme weather detection and analytics. Documented data workflows and system architecture for reproducibility.

Spatial Data Analysis 2025
Implemented an R-Tree index for efficient spatial querying, developing kNN and skyline algorithms for location-based analysis. Benchmarked multiple algorithms for performance and scalability. Presentation: Google Drive link

Bitcoin Price Prediction at Databricks (Certificate link) 2025
Performed exploratory data analysis (EDA) and visualized Bitcoin market trends, ingesting data via REST API and preprocessing datasets for machine learning pipelines. Implemented a regression model in Databricks to predict Bitcoin price movements.

Flight Delay Analysis System 2023
Analyzed delay patterns using Pandas and built interactive visualizations in Bokeh, developing a recommendation model for airline selection based on delay trends. Built interactive dashboards to support data exploration.

Machine Learning & NLP Projects

Graph Neural Network Replication and Generalisation Study 2026
Reproduced the Graph Isomorphism Network (GIN, ICLR 2019) on the IMDB benchmarks within the paper's reported error, then tested generalisation on new graph datasets scraped from TMDb. Ran training jobs on a university GPU server.

Multilingual ED Triage Translation 2025
Designed safe Chinese↔English translation for clinical Limited English Proficiency (LEP) scenarios, processing medical datasets and building tokenization and validation pipelines. Fine-tuned NLLB-200 using GPU training with Hugging Face workflows and evaluated models using BLEU/chrF.